

## SHREYA CHAKRABORTY

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### Education

Master in Business Analytics & Information Systems, University of South Florida, Tampa, Florida, United States CGPA 3.81  
Bachelors in Electronics and Communication Engineering, Manipal Institute of Technology, Manipal, Karnataka, India CGPA 3

### Work Experience

#### Intern at ELAS, USA

July 2025 – Present

- Designed and implemented an AI-powered HR recruitment agent using LangChain that automates job posting, custom question generation, candidate response collection, and scoring using structured JSON-based workflows and RESTful API endpoints.
- Built modular backend architecture with endpoints for posting job descriptions, creating role-specific questions, retrieving interview forms, submitting responses, and rating candidates ensuring a seamless flow between recruiter and candidate interactions.
- Engineered persistent data storage system using uniquely identified JSON files, allowing easy tracking of job roles, candidate responses, and final ratings for informed hiring decisions

#### Data Analyst, University of South Florida, Muma College of Business, Tampa, Florida, USA

Feb 2024 – May 2025

- Developed Python and ML scripts to automate certificate issuance and detect duplicate records, streamlining administrative processes, providing 100% improvement in accuracy and 40% improvement in total process time. Maintained a master SQL database for the course, ensuring organized and accurate data management.
- As a Student Mentor at the University of South Florida, guided undergraduate students through various data visualization projects as part of the Citizen Data Science Certificate program launched in collaboration with Tableau at the Muma College of Business. Delivered presentations to students to introduce the goals and structure of the program, helping them understand how to apply data analysis techniques using Tableau in real-world contexts. Projects I supported included a consumer sentiment analysis for Cigar City Brewing (CCB), where students worked with over 75,000 BeerAdvocate reviews to explore trends across beer styles, ABV levels, and sensory attributes, and a stock market analysis project where students used financial data to build regression models, forecast stock prices, and visualize statistical indicators like average and standard deviation. In each case, I helped students create calculated fields, build interactive dashboards, and develop compelling data-driven stories, ultimately strengthening their analytical thinking and Tableau proficiency.

#### Data Engineer, Accenture, Bangalore, India

May 2021 – May 2023

- Designed and implemented an end-to-end automated data pipeline using **Automation Anywhere and Python** to process large datasets (128 columns, 1000+ rows per file), streamlining the creation of **100,000 sales and acquisition forms in 48 hours**
- Built robust **ETL logic to ingest, transform, and validate** structured datasets before integration with Accenture's internal CRM application (*Manage My Sales*). Developed an ETL pipeline using Airflow to automate the extraction of economic time-series data from the FRED API for multiple indicators (GDP, CPI, etc.). Implemented data cleaning and preprocessing logic, storing intermediate series objects using Python pickle files for efficient caching and reuse. Designed automated database load steps that used UPSERT PL/SQL procedures to insert or update data in Oracle databases, ensuring no duplication using dynamic bookmark logic. Monitored the entire workflow through Airflow DAGs, enabling scheduled data refresh, logging start/stop timestamps, and ensuring reliability for academic data warehousing projects.
- Developed and deployed a full-stack Flask web application using the "MovieLens" dataset to allow users to rate movies and receive personalized recommendations using cosine similarity algorithm. Implemented content-based filtering to generate dynamic suggestions based on user preferences, enhancing engagement and usability. Integrated an Oracle database to manage user data, movie metadata, and recommendation history. Utilized stored procedures for efficient rating insertion and used secure user authentication.

#### Project Intern - Biz4Solutions, Pune, India

Dec 2020- May 2021

- Implemented algorithms using python to predict revision time intervals personalized for a student based on his test score history
- Test results of 10,000 students over 12 test cycles were used to build the algorithms. Prediction accuracy was 100% on a sample size of 100 students. To manage the large dataset (2 TB), Mongo DB in AWS environment was used

### Projects During Masters Program

#### Predicting the success of bank telemarketing using data modelling

- Developed a machine learning model to predict bank telemarketing success, analysing 45,000 client records
- Applied **AI/ML models XGBoost, SVM, Random Forest, Neural Networks, Logistic Regression, and Decision Tree**, used **hyperparameter tuning** to improve performance. **Logistic Regression achieved the highest recall score of 93%**, ensuring effective identification of potential subscribers

#### Big Data Processing & Predictive Analytics using PySpark

- Study of Racial Bias in Police Stops (Nashville, TN): Analysed 3.2M+ police stop records from the Stanford Open Policing Project using PySpark for data ingestion, cleaning, and transformation. Addressed missing values, standardized column names, and stored the processed data in a persistent table for analysis.
- Conducted Exploratory Data Analysis (EDA) to examine arrests by race, gender, and age. Identified key patterns, including racial disparities in top states and a moderate correlation between age and arrest likelihood.
- Tackled big data challenges such as memory management in a distributed environment while handling large datasets in Spark.
- Developed insights that will contribute to a binary prediction model using Spark ML for identifying potential biases in policing **outcomes**.

### Skills

**Cloud Platforms** - AWS (S3, Lambda, DynamoDB, Athena, SQS, EC2, RDS), Azure, Google Cloud, Cloud Architecture

**AI/ML** - Deep Learning Models (Transformers, CNNs, RNNs), Self-Supervised Learning, Zero-shot Models, Prompt Engineering, Hugging Face Transformers  
Fine-tuning LLMs using PEFT, LoRA, QLoRA, and RAG with LangChain, Text Processing (TF-IDF, Word2Vec, BOW), Elasticsearch  
Feature extraction (POS tagging, NER), Unsupervised Learning (clustering, data modelling, dimensionality reduction)

**Frameworks** - Spring Boot, Django, Flask, Entity Framework, ASP.NET, CI/CD, Power BI, Anaconda

**Programming Languages** - Python, Java, C#, C++, SQL, HTML, CSS, JavaScript, ASP.net MVC, Dot Net, REST API, Express, Node

**Big Data Technology** - Mongo DB, Hadoop, Spark, MapReduce, Apache Airflow, Snowflake, Oracle, monitoring, data quality, reporting, dashboards

**Version Control and Database** - GitHub, Git, SQL Server, PostgreSQL, NoSQL, Cassandra

**Libraries** - TensorFlow, Scikit-learn, NLTK, Hugging Face Transformers, Pandas, Numpy, Matplotlib, Pytorch

**Tools** - UiPath, Automation Anywhere, Tableau, Docker, CI/CD, Agile, Microsoft Excel, Microsoft Power Point, Microsoft Word, Data modelling, Jupyter

### Awards & Recognition

- Published a paper in International Journal of Engineering Research and Technology titled "Continuous Variable Transmission using Raspberry Pi and Stepper Motor" in IJERT, Volume 11, Issue 2, Feb 2022.
- Received 'Sparkling Star Award' and client appreciation at Accenture in 2022

<https://scsilvershark.wixsite.com/my-site-1/portfolio-1>, <https://www.linkedin.com/in/shreya-chakraborty-5839b5189/>, <https://github.com/Shreya-tech1998>